POLICY BRIEF | PLFS 2021–22 DATA ANALYSIS

# BEYOND ECONOMIC GROWTH

Why Macroeconomic Indicators Fail to Explain Female Labour Force Participation in India.


## EXECUTIVE SUMMARY

Conventional economic planning operates on the assumption that macroeconomic growth (NSDP), urbanisation, and human capital investments (secondary education) automatically translate into higher workforce participation for women. Analysis of PLFS 2021–22 state-level data demonstrates that these traditional structural proxies hold virtually zero statistically significant explanatory power over Female Labour Force Participation (FLFPR). In multivariable OLS specifications, individual predictors such as secondary education ($p = 0.944$), log NSDP ($p = 0.402$), and urbanisation ($p = 0.145$) fail to show significance. To unlock female economic agency, state and national policies must shift away from aggregate macro-growth strategies and directly address the latent, micro-structural realities of unpaid care burdens, safety deficits, and rigid socio-cultural norms.


### 1. The Empirical Paradox

Across 33 Indian states/UTs, standard socioeconomic metrics yield low predictive power and un-intuitive directions of association:

- **Education Paradox ($R^2 = 0.013$, $p = 0.944$):** Secondary education completion shows no positive linear relationship with FLFPR, as education alone cannot override job quality deficits or cultural constraints.
- **Urbanisation Mismatch ($r = -0.404$):** Urbanisation exhibits a negative correlation with female participation. Structural shifts away from agriculture shrink rural informal roles without expanding accessible urban employment.
- **Economic Growth Divergence ($p = 0.402$):** Log NSDP fails to predict FLFPR. Capital-intensive state growth increases output without generating proportional female job opportunities.

### 2. Academic Literature Synthesis

The statistically insignificant empirical coefficients reflect documented structural mechanisms in labor economics literature:

**The Feminization U-Curve & Income Effect**

As established by Goldin (1995) and Klasen & Pieters (2015) for India, rising household incomes during early growth lead to an 'income effect' where women exit manual labor, while occupational segregation limits their formal white-collar entry.

**The Education-Aspiration Mismatch**

Fletcher, Pande, & Moore (2017) demonstrate that while female schooling expanded rapidly, formal service/manufacturing job creation lagged. Moderately educated women face high opportunity costs and social stigma in informal manual work, resulting in exit.

**Model Performance Summary: Standard Macroeconomic Predictors**

| Specification Model | Sample (N) | $R^2$ | Adjusted $R^2$ | Key Statistical Observation |
|---|---|---|---|---|
| Education Only | 33 | 0.013 | -0.019 | Statistically insignificant slope ($p = 0.524$) |
| Economic Output (Log NSDP) | 32 | 0.043 | 0.012 | No significant association with FLFPR ($p = 0.253$) |
| Demand-Side (Urban + Non-Agri) | 33 | 0.175 | 0.120 | Urbanisation ($p = 0.375$), Non-Agri ($p = 0.512$) |
| Full Multivariable Model | 32 | 0.405 | **0.262** | All 6 individual predictors statistically insignificant |

*Source: Empirical OLS estimations from PLFS 2021–22 dataset state-level aggregations.*

### 3. Structural Outliers & Qualitative Latent Drivers

### The Meghalaya Insight: Norms vs. NSDP

Regression diagnostics identify Meghalaya as an extreme influential structural outlier with a Cook's Distance of $D_i = 1.047$ (over 8 times the leverage threshold of $4/N = 0.125$). Meghalaya sustains exceptionally high female labor participation despite modest urbanization and economic output.

**Key Policy Insight:** Matrilineal kinship structures and equitable social norms exert far stronger structural influence on female workforce participation than macroeconomic performance. Aligning institutional care and social acceptability unlocks labor supply regardless of economic output.

### What Standard Macro Models Omit

State-level aggregations mask micro-level barriers:

- **Unpaid Care Burden:** NSO Time Use Surveys show Indian women spend 299 mins/day on domestic care vs. 25 mins for men.
- **Spatial & Safety Deficits:** Inadequate transit infrastructure severely restricts safe mobility in urban clusters.
- **Formal Service Deficit:** Absence of flexible, middle-skill formal jobs suitable for educated women.

## 4. Actionable & Innovative Policy Solutions

Since economic growth alone will not spontaneously raise female participation, state governments must deploy targeted interventions designed to dismantle institutional friction and micro-structural barriers.

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| **A. Care-Economy Infrastructure**<br>Care-as-Infrastructure Grants: Mandate and co-fund decentralized, neighborhood-level creches (under expanded Palna schemes) co-located near industrial corridors, commercial zones, and urban informal clusters to directly reduce time poverty for working mothers. | **B. Safe Mobility Interventions**<br>Gender-Informed Urban Transit: Implement 'Pink Transit Express' routes, last-mile electric feeder connectivity, and subsidized fare programs (e.g., Karnataka's Shakti scheme) combined with lighting audits to unlock safe urban transit access. |
| **C. White-Collar Apprenticeships**<br>Targeted Service Incentives: Provide tax credits to IT-enabled services, healthcare, social commerce, and logistics firms that establish hub-and-spoke operational centers in Tier-2/3 towns, creating localized white-collar roles for educated women. | **D. Innovative Micro-Policy**<br>Social Security for Gig & Homeworkers: Establish state portable benefit boards for home-based informal workers and female gig platform workers. Provide micro-grants for home-office setups and digital micro-enterprises to combine home care and wage earning. |

*Drafted based on PLFS 2021–22 Econometric Analysis | Author: Sivachanthini Ravikumar*